90

1. 20
2. 30
3. ~~10~~ 9
4. 20

MATA KULIAH UMUM
POLITEKNIK NEGERI BANDUNG

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PROGRAM STUDI : D4 Telekomunikasi Nirkabel
MATA PELAJARAN : Matematika Terapan I
DOSEN : Endang H.
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[illegible]

بِسْمِ اللَّهِ الرَّحْمَٰنِ الرَّحِيمِ

1. $Y = 4\cos x + \sqrt{48}\sin x$, dengan x dlm degree (radian)

a. Ubahlah dalam bentuk $Y_A = R\sin(x-\alpha)$

$\Rightarrow 4\cos x + \sqrt{48}\sin x = R\sin(x-\alpha)$

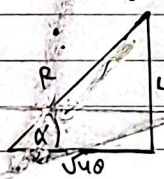
$4\cos x + \sqrt{48}\sin x = R\sin x\cos\alpha - R\cos x\sin\alpha$

$\Rightarrow 4\cos x = -R\cos x\sin\alpha$

$\sin\alpha = -\frac{4}{R}$

$\Rightarrow \sqrt{48}\sin x = R\sin x\cos\alpha$

$\cos\alpha = \frac{\sqrt{48}}{R}$



$R = \sqrt{(4)^2 + (\sqrt{48})^2}$
$= \sqrt{16+48}$
$= \sqrt{64}$
$= 8$

$\tan\alpha = \frac{\sin\alpha}{\cos\alpha} = \frac{-4/R}{\sqrt{48}/R} = -\frac{4}{4\sqrt{3}}$

tan Kuadran IV $\longrightarrow \alpha = \tan^{-1}\left(\frac{-1}{\sqrt{3}}\right)$
$= -30^\circ$
$= -0{,}523599$

Jadi, $Y = 4\cos x + \sqrt{48}\sin x = 8\sin(x-(-30^\circ))$
$= 8\sin(x+30^\circ)$
$= 8\sin(x+0{,}523599)$

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b. Tent. 2 buah x jika $y = 4\sqrt{3}$

$Y = 8\sin(x+0{,}523599)$

$4\sqrt{3} = 8\sin(x+0{,}523599)$

$\sin^{-1}\left(\frac{4\sqrt{3}}{8}\right) = x + 0{,}523599$

$\frac{\pi}{3} + k\cdot 2\pi = x + 0{,}523599$

$k=0 \Rightarrow \frac{\pi}{3} = x + 0{,}523599$

$= 0{,}52356$

$\pi - \frac{\pi}{3} + k\cdot 2\pi = x + 0{,}523599$

$k=0 \Rightarrow \frac{2\pi}{3} = x + 0{,}523599$

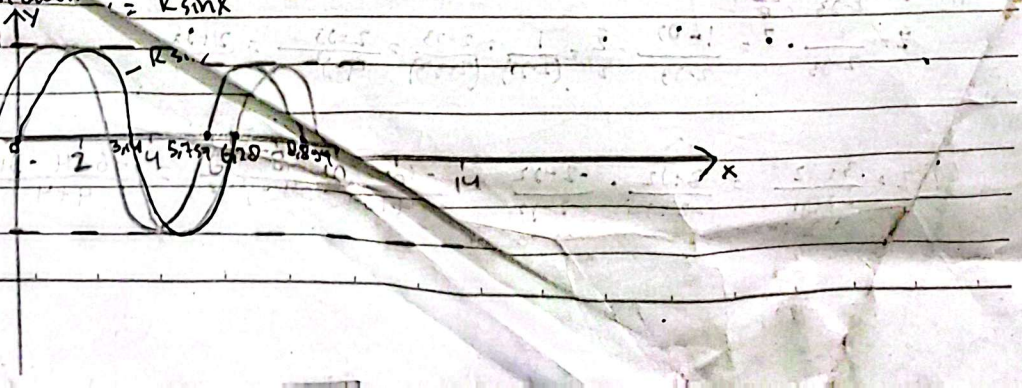
$x = 1{,}57079$

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C. Sketsa grafik y dgn acuan $= R\sin x$

$T = \frac{2\pi}{\omega} = \frac{2\pi}{1} = 2\pi$
$= 6{,}28_s$

$\text{phase} = \frac{\alpha}{\omega}$
$= \frac{0{,}523599}{1}$
$= 0{,}523599$



2. $V = 8\sin\omega t + 6\cos\omega t$ volt. Jika $\omega = 3$ rad/sec, t dlm sec,

a. Ubahlah/nyatakan V dalam bentuk $R\cos(\omega t + \alpha)$

$V = 8\sin\omega t + 6\cos\omega t = R\cos(\omega t + \alpha)$

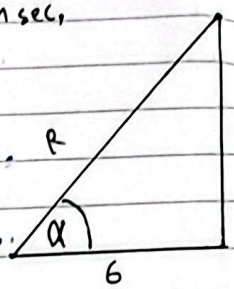
$8\sin\omega t + 6\cos\omega t = R\cos\omega t\cos\alpha - R\sin\omega t\sin\alpha$

$\Rightarrow 8\sin\omega t = -R\sin\omega t\sin\alpha$

$\sin\alpha = -\frac{8}{R}$

$\Rightarrow 6\cos\omega t = R\cos\omega t\cos\alpha$

$\cos\alpha = \frac{6}{R}$



$R = \sqrt{(6)^2 + 8^2}$

$= \sqrt{36+64}$

$= \sqrt{100}$

$= 10$

$\tan\alpha = \frac{\sin\alpha}{\cos\alpha}$

$= \frac{-8/R}{6/R} = -\frac{4}{3}$

tan di kuadran IV $\rightarrow \alpha = \tan^{-1}(-\frac{4}{3})$

$= -0,92729$

Jadi $V = 10\cos(\omega t + (-0,92729)$

$= 10\cos(\omega t - 0,92729)$

$= 10\cos(\omega t - 53,13°)$

$= 10\cos(3t - 53,13°)$

b. Berapa V pada $t = 4$ second

$V = 10\cos(3t - 0,92729)$

$V = 10\cos(3(4) - 0,92729)$

$= 10\cos(11,07271)$

$= 0,7706$ m/s

c. Kapan t bila $V = 5$ volt

$V = 10\cos(3t - 0,92729)$

$5 = 10\cos(3t - 0,92729)$

$\cos^{-1}(\frac{5}{10}) = 3t - 0,92729$

$1,04719 = 3t - 0,92729$

$t = \frac{1,04719 + 0,92729}{3} = 0,6585$

d. Berapa perbedaan waktu V (tekan lagging atau leading) $V_2 = 6\cos 3t$

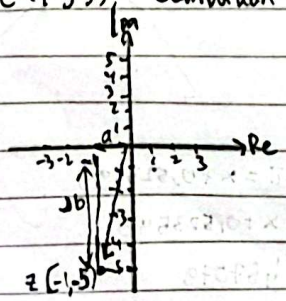
$V = 10\cos(3t - 0,92729)$ → grafik geser ke kanan sejauh $0,92729_5$

phase $= \frac{\alpha}{\omega} = \frac{0,92729}{3} = 0,30909$

Jadi, $V = 10\cos(3t - 0,92729)$ lagging thd $V_2 = 6\cos 3t$ sebesar $0,30909_5$

3. a) $z = (3 - j2) - (4 + j3)$ Gambarkan z dalam bidang kompleks

$z = (-1 - j5)$



b) $z = \frac{1}{2-j3}$

$z = \frac{1}{2-j3} = \frac{1-0j}{2-j3} \cdot \frac{z}{z} = \frac{1}{(2-j3)} \cdot \frac{2+j3}{(2+j3)} = \frac{2+j3}{4-j^2} = \frac{2+j3}{5}$

$\frac{2+j3}{13} = \frac{2}{13} + \ldots$

c) $z = \frac{3+j2}{-2+j2} = \frac{3+j2}{-2+j2} \cdot \frac{-2-j2}{-2-j2} = \frac{-6-j6-j4-j^2 4}{4+j4-j4-j^2 4} = \frac{-6+4-j10}{4+4} = \frac{-2-j10}{8} = \frac{-1}{4} - \frac{j10}{8}$

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4. $z_1 = -3+j2$ $\quad z_2 = 2-j3$

a) $z_1 . z_2$ b) $\frac{z_2}{z_1}$ Hitung dlm polar

$z_1 = -3+j2$ → kuadran 2

$r = \sqrt{(-3)^2 + (2)^2}$

$= \sqrt{9+4} = \sqrt{13}$

$\theta = \tan^{-1}\left(\frac{b}{a}\right) =$

$= \tan^{-1}\left(\frac{2}{-3}\right)$

$= -0{,}588$ → K2

$= 146{,}309°$

$z_1 = \sqrt{13} \angle 146{,}309°$

$z_2 = 2-j3$ → kuadran 4

$r = \sqrt{(2)^2 + (-3)^2}$

$= \sqrt{4+9}$

$= \sqrt{13}$

$\theta = \tan^{-1}\left(\frac{b}{a}\right)$

$= \tan^{-1}\left(\frac{-3}{2}\right)$ kuadran 4 $= -0{,}9827$

$= 303{,}69°$

$z_2 = \sqrt{13} \angle 303{,}69°$

a. $z_1 . z_2 = \left(\sqrt{13} \angle 146{,}309°\right)\left(\sqrt{13} \angle 303{,}69°\right)$

$= 13 \angle 449{,}999°$

b. $\frac{z_2}{z_1} = \frac{\sqrt{13} \angle 303{,}69°}{\sqrt{13} \angle 146{,}309°} = 1 \angle 157{,}381°$

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